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ANALYSING THE BEHAVIOUR OF NETWORK DEVICES USING CISCO PACKET TRACER SIMULATOR

AIM:

To analyse the behaviour of network devices such as hubs and switches using Cisco Packet Tracer and study how they forward data packets between connected computers.

ALGORITHM:

1. Open Cisco Packet Tracer on the computer.
2. Place four PCs and one hub in the logical workspace.
3. Connect all four PCs to the hub using Copper Straight-Through cables.
4. Assign IP addresses and subnet masks to all the PCs.
5. Place another four PCs and one switch in the workspace.
6. Connect all four PCs to the switch using Copper Straight-Through cables.
7. Assign IP addresses and subnet masks to the PCs connected to the switch.
8. Select the Add Simple PDU tool from the common tools bar.
9. Click the source PC and then the destination PC to send a packet.
10. Observe the packet transmission in Realtime or Simulation mode.
11. Repeat the procedure for both the hub-based and switch-based networks.
12. Compare the packet-forwarding behaviour of the hub and switch.

IP ADDRESS CONFIGURATION:

Hub Network:

PC0 – 192.168.1.1

PC1 – 192.168.1.2

PC2 – 192.168.1.3

PC3 – 192.168.1.4

Subnet Mask – 255.255.255.0

Switch Network:

PC4 – 192.168.2.1

PC5 – 192.168.2.2

PC6 – 192.168.2.3

PC7 – 192.168.2.4

Subnet Mask – 255.255.255.0

OUTPUT:

1. Hub: When a packet is received from a PC, the hub forwards it to all other connected ports except the port from which it was received.

2. Switch: When a packet is received, the switch learns the MAC addresses of connected devices and forwards the frame to the appropriate destination port when the destination is known.

3. Packet transmission was simulated successfully between the source and destination PCs in both networks.

RESULT:

Thus, the behaviour of network devices was analysed using Cisco Packet Tracer. The experiment demonstrated that a hub broadcasts incoming data to all other ports, whereas a switch forwards data selectively based on MAC address information, reducing unnecessary network traffic.